

HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

COMP 3421 WEB APPLICATION DESIGN AND DEVELOPMENT

MID-TERM EXAMINATION (ANSWER SHEET)

28 OCTOBER 2025

NAME: _____

STUDENT ID: _____

Instructions:

1. The time allowed is **2** hours.
2. This examination has three sections. **Answer ALL the questions on the answer sheet.**
3. Marks for each question are shown next to the question.

SECTION A (30 MARKS)	SECTION B (40 MARKS)	SECTION C (30 MARKS)	TOTAL

SECTION A (30 MARKS)

(1) B	(2) A	(3) B	(4) C	(5) D
(6) C	(7) B	(8) B	(9) B	(10) B
(11) B	(12) A	(13) D	(14) B	(15) B

SECTION B — SHORT QUESTIONS (40 MARKS)

Q1	Inline CSS is written directly in an HTML element's style attribute, affecting only that element. Embedded CSS is placed inside a <style> block in the HTML document's <head>, applying to the whole page. External CSS is stored in a separate .css file linked via <link>, allowing reuse across multiple pages.
Q2	<pre>h2 { text-transform: uppercase; letter-spacing: 10px; text-align: center; }</pre>
Q3	<pre>body { background-image: url("pattern.png"); background-repeat: repeat-x; background-attachment: fixed; }</pre>

Q4	"Times New Roman" is enclosed in quotation marks because it contains spaces; multi-word font names must be quoted in CSS to be recognized as a single font family name.
Q5	The <code>alt</code> attribute provides alternative text for an image if it cannot be displayed, used by screen readers and for SEO.
Q6	"Cascading" refers to the way CSS resolves conflicts when multiple rules apply to the same element, based on source order, specificity, and importance.
Q7	The border is part of the box model, affecting layout and dimensions; the outline is drawn outside the border, does not take up space, and is often used for focus indicators.

Q8	<pre>.box { width: 300px; height: 200px; border: 5px solid red; padding: 20px; }</pre>
Q9	Block-level elements take the full width of their container and start on a new line (e.g., <code><div></code>), while inline elements take only necessary width and stay in the flow of text (e.g., <code></code>).
Q10	<pre><video controls> <source src="promo.mp4" type="video/mp4"> </video></pre>

SECTION C — LONG QUESTIONS**QUESTION 1 (10 MARKS)**

```


<map name="polyuMap">

  <area
    shape="rect" coords="350,300,450,350"
    href="http://comp.polyu.edu.hk"
  />

  <area
    shape="rect" coords="170,260,370,360"
    href="http://library.polyu.edu.hk"
  />

  <area
    shape="circle" coords="325,255,50"
    href="http://hro.polyu.edu.hk"
  />
</map>
```

SECTION C — LONG QUESTIONS**QUESTION 2 (10 MARKS)**

The selector `:not(p)` means “**select every element that is NOT a `<p>` element.**”

So, this rule applies `color: red;` to **all elements except `<p>`.**

SECTION B — LONG QUESTIONS**QUESTION 3 (10 MARKS)**

border-radius: 70% 30% 30% 70% / 60% 40% 60% 40%;

****END****